Q5

$\psi = 2.0$

For attr. X

unit	density
X1	0
X2	0.4
X3	0.4
X4	0.2

$H(X) = -0 \log 0 - 0.4 \log 0.4 - 0.4 \log 0.4 - 0.2 \log 0.2$

$= 1.52$

For attr. Y

unit	density
Y1	0.2
Y2	0.4
Y3	0
Y4	0.4

$H(Y) = -0.2 \log 0.2 - 0.4 \log 0.4 - 0 \log 0 - 0.4 \log 0.4$

$= 1.52$

For attr. Z

unit	density
Z1	0.4
Z2	0
Z3	0.4
Z4	0.2

$H(Z) = -0.4 \log 0.4 - 0 \log 0 - 0.4 \log 0.4 - 0.2 \log 0.2$

$= 1.52$

$L_1 = \{ \{X\}, \{Y\}, \{Z\} \}$

Size 2 subspace Generation

Candidate Generation

Join / Prune Step:

$C_2 = \{ (X, Y), (X, Z), (Y, Z) \}$

Good Subspace Generation

Country Step:

$L_2 = \{$ $\}$

XY:

	Y1	Y2	Y3	Y4
X1	0	0	0	0
X2	0	0.4	0	0
X3	0	0	0	0.4
X4	0.2	0	0	0

$H(X,Y) = -0.4 \log 0.4 - 0.4 \log 0.4 - 0.2 \log 0.2 = 1.52$

XZ:

	Z1	Z2	Z3	Z4
X1	0	0	0	0
X2	0.1	0	0.2	0.1
X3	0.1	0	0.2	0.1
X4	0.2	0	0	0

$H(X,Z) = -(0.1 \log 0.1) \times 4 = (0.2 \log 0.2) \times 3 = 2.72$

YZ

	Z1	Z2	Z3	Z4
Y1	0.2	0	0	0
Y2	0.1	0	0.2	0.1
Y3	0	0	0	0
Y4	0.1	0	0.2	0.1

$H(Y,Z) = -(0.1 \log 0.1) \times 4 - (0.2 \log 0.2) \times 3 = 2.72$

$L_2 = \{ \{X,Y\} \}$

Size 3-subspace Generation

Candidate Generation

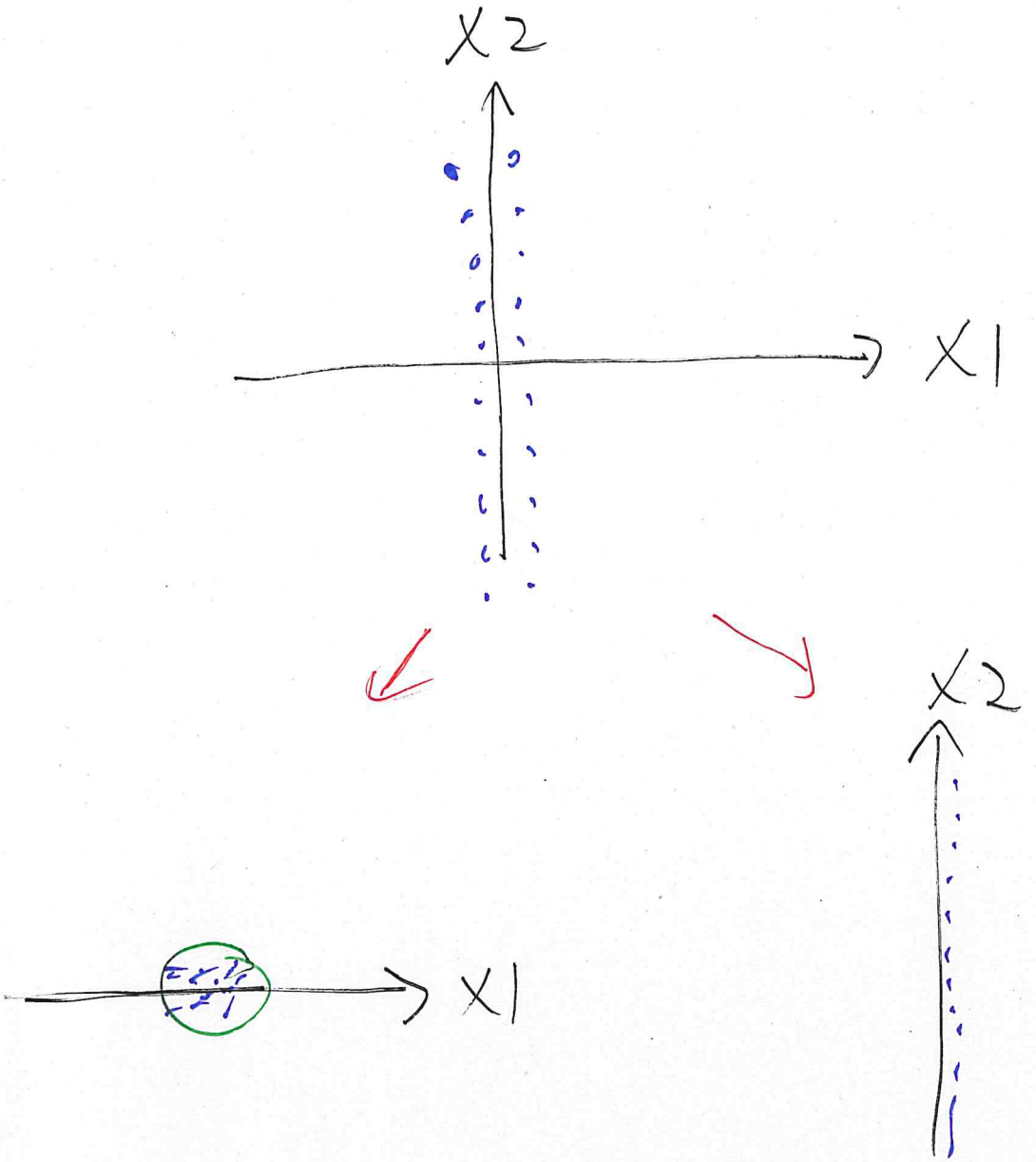
Join Step

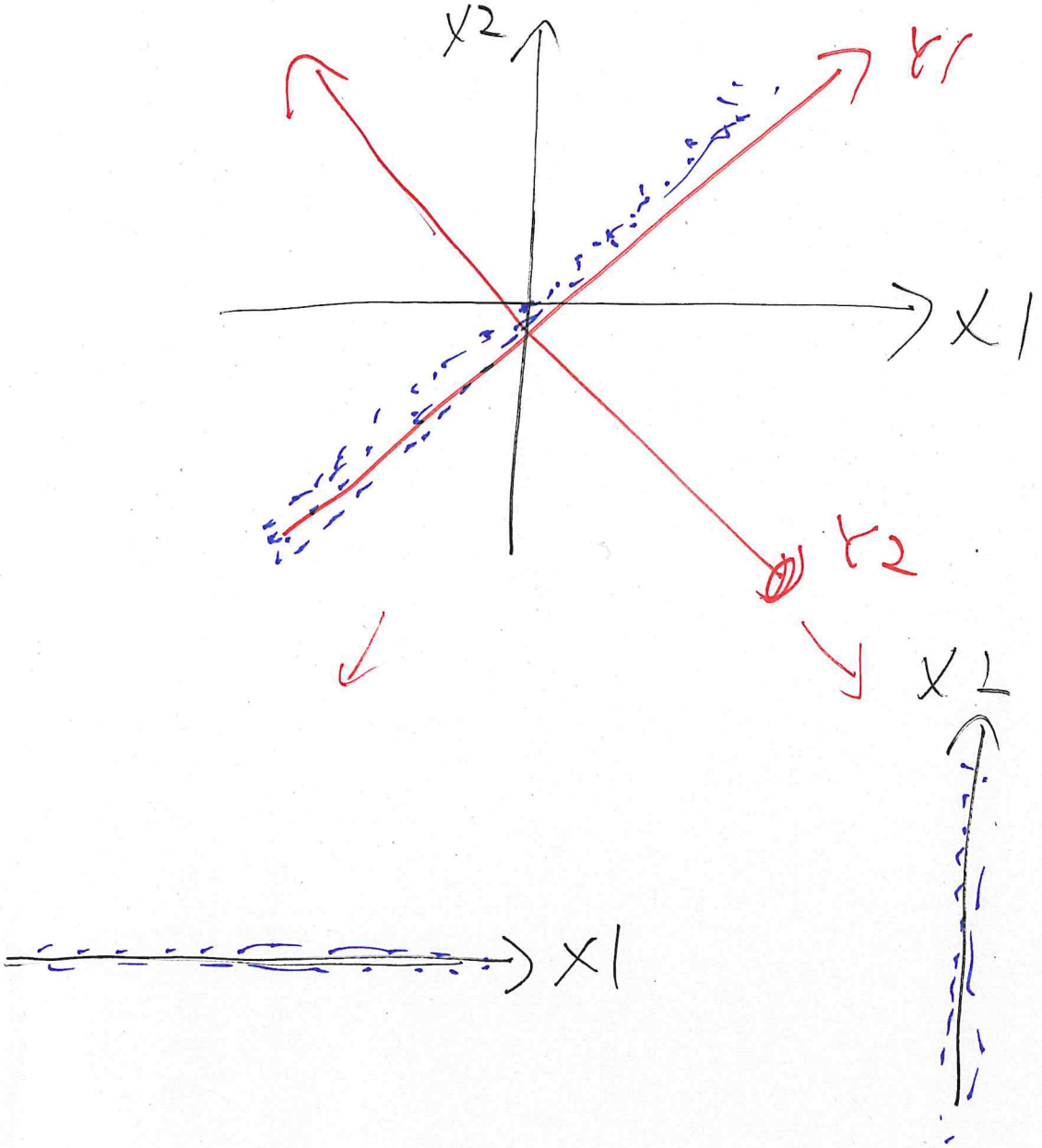
$C_3 = [\ ]$

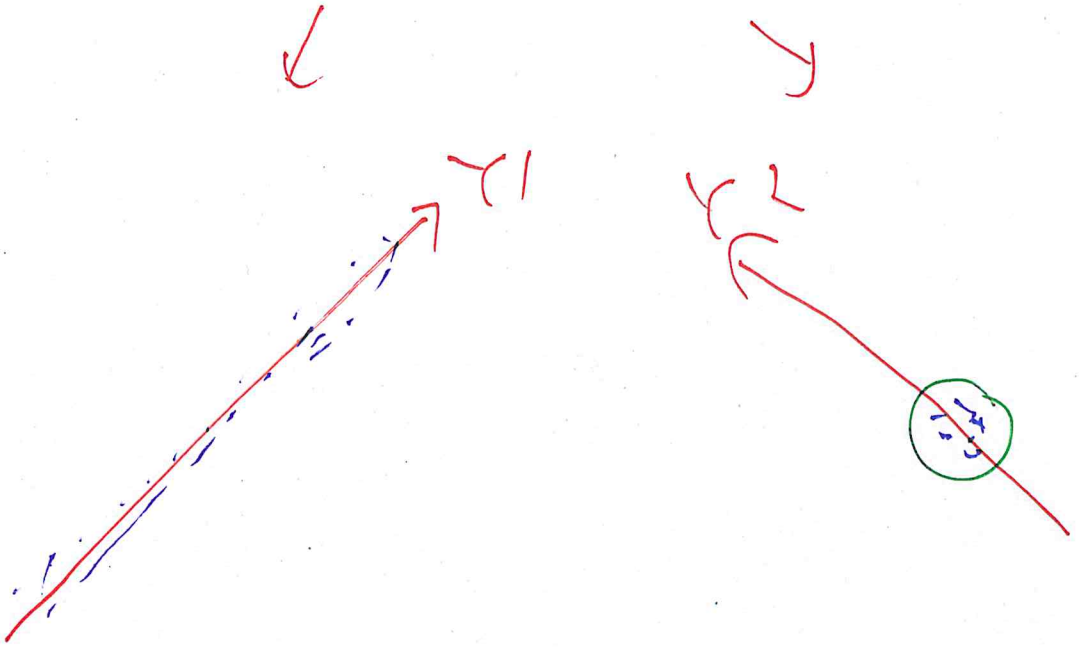
subspaces containing good clusters

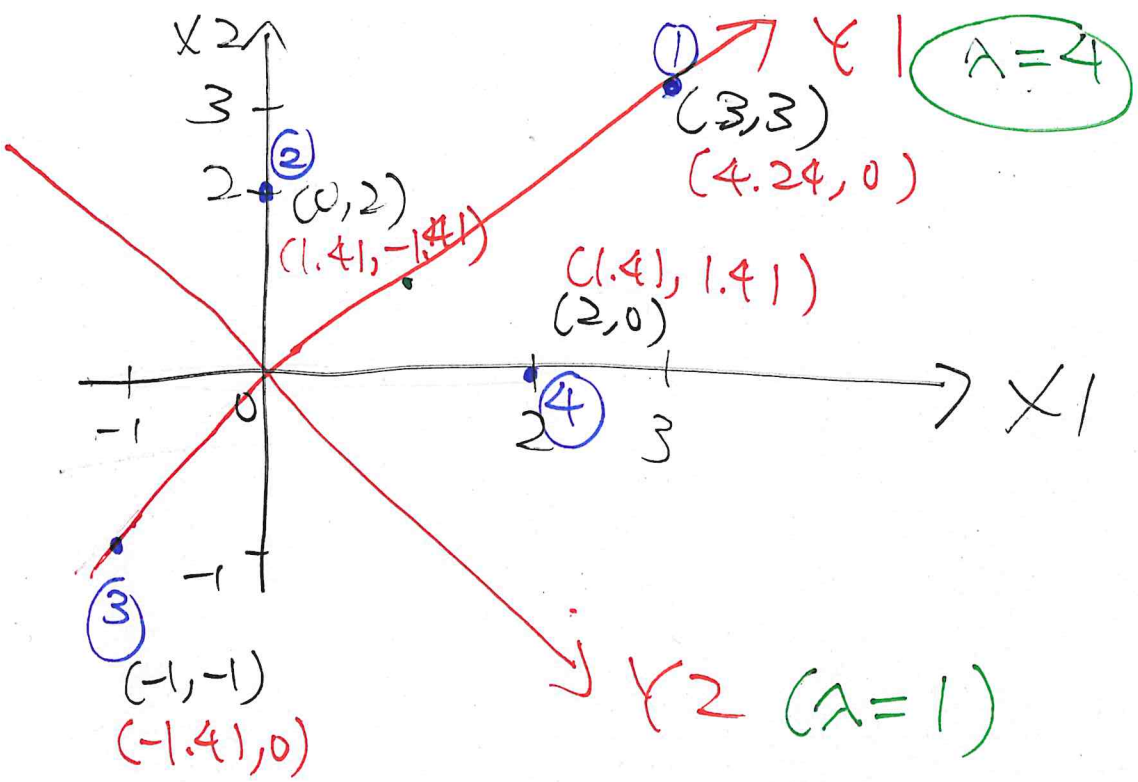
$= \{ [X], [Y], [Z], [X,Y] \}$

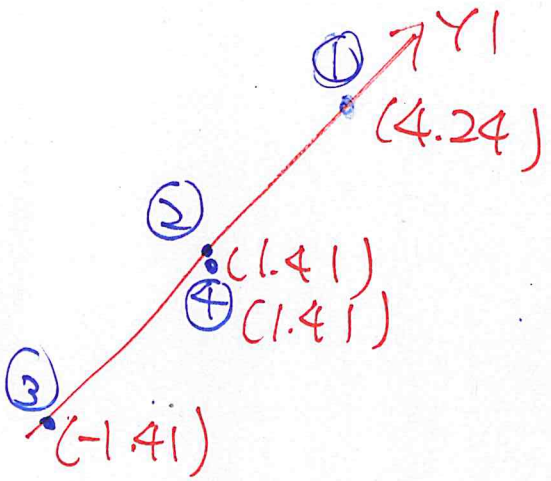
	dense unit	entropy
Adv	- can find clusters (w.r.t subspaces) directly	efficient if we are only interested in finding subspaces only (without clusters)
Disadv	inefficient if we are only interested in finding subspaces only (without clusters)	- cannot find clusters (w.r.t subspaces) directly

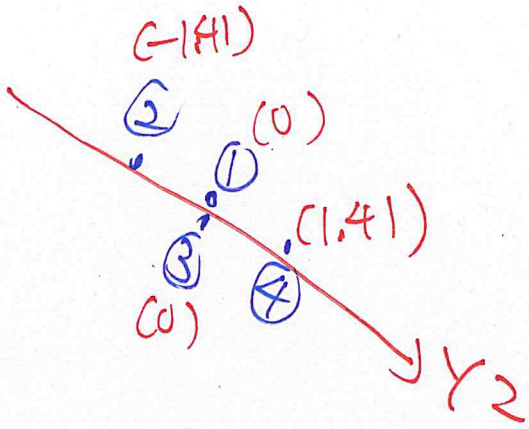

X2
X1
X2
X1


X2
Y1
X1
Y2
X1
X1




X2
3
2
-1
0
-1
2
3
X1
Y1
λ=4
(3,3)
(4.24, 0)
(0,2)
(1.41, -1.41)
(1.41, 1.41)
(2,0)
(-1,-1)
(-1.41, 0)
Y2 (λ=1)


Y1
(4.24)
(1.41)
(1.41)
(-1.41)

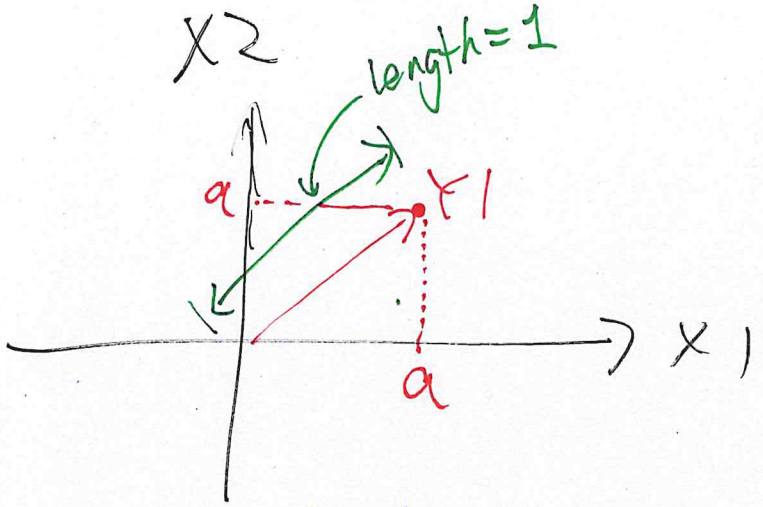

(-1.41)
(0)
(1.41)
(0)
Y2

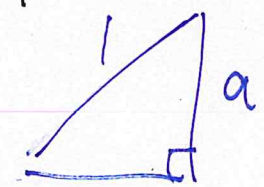
x2
x1
z1
z2

z1
z2

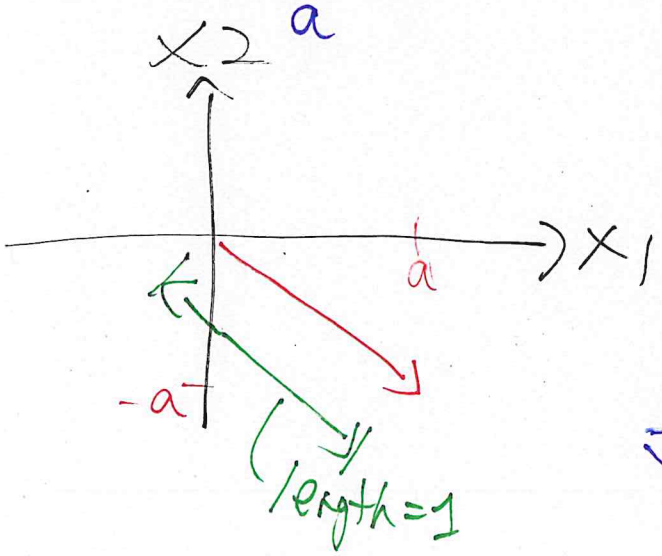
$$\begin{cases} -\frac{3}{2}x_1 + \frac{3}{2}x_2 = 0 \\ \frac{3}{2}x_1 - \frac{3}{2}x_2 = 0 \end{cases} \quad \times\frac{2}{3} \quad \times\frac{2}{3}$$

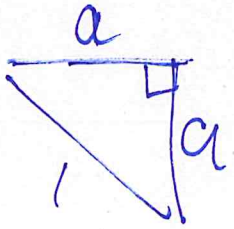
$$\begin{cases} -x_1 + x_2 = 0 \\ x_1 - x_2 = 0 \end{cases}$$





$$a^2 + a^2 = 1^2$$
$$2a^2 = 1$$
$$a^2 = \frac{1}{2}$$
$$a = \sqrt{\frac{1}{2}}$$





$$a = \sqrt{\frac{1}{2}}$$